

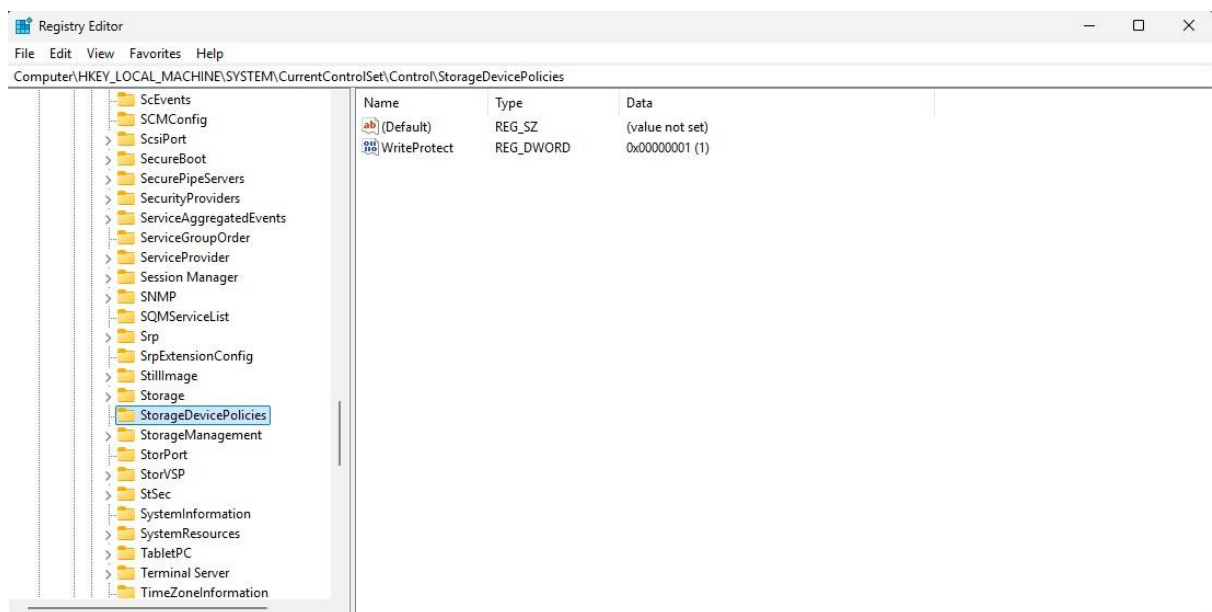
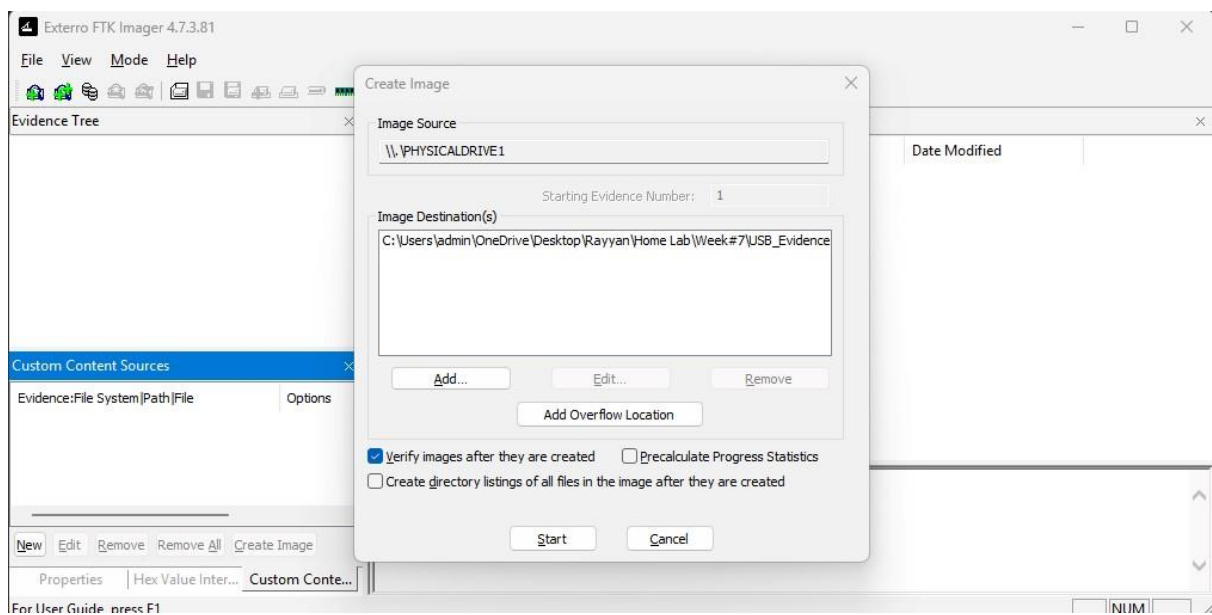


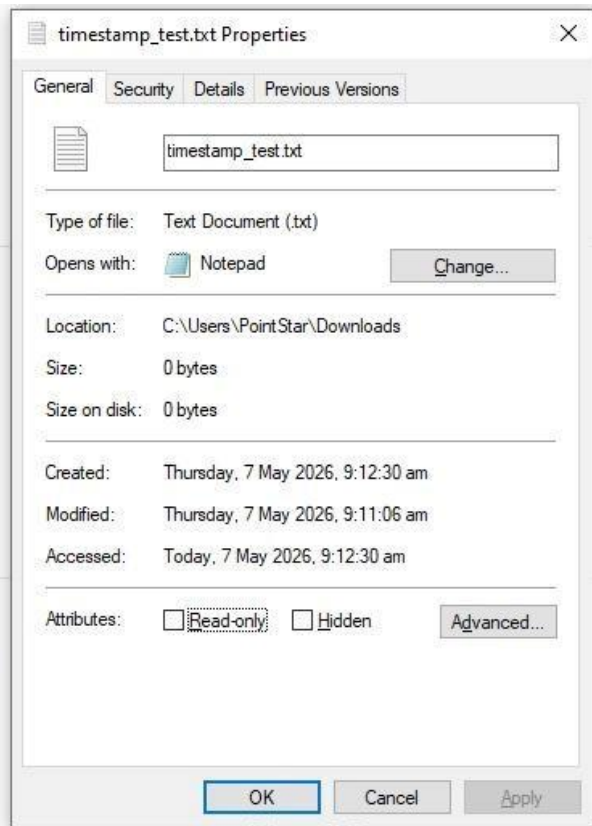
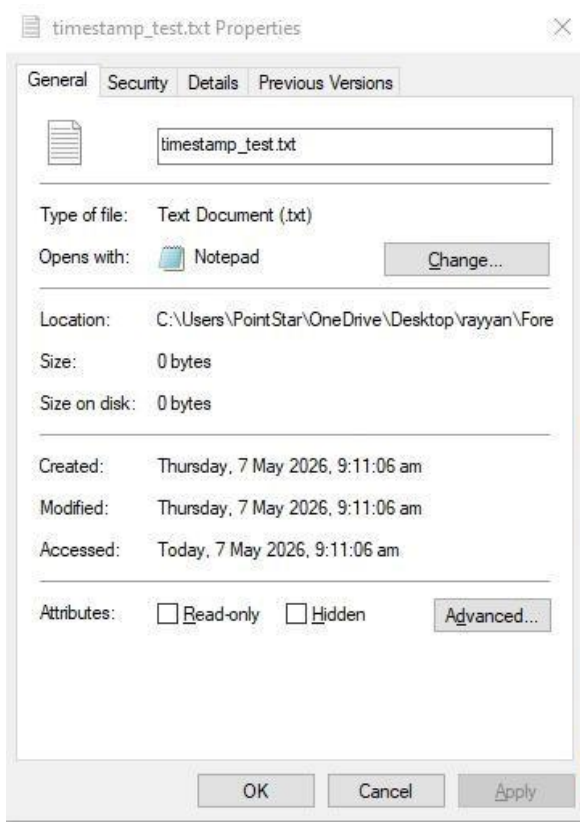
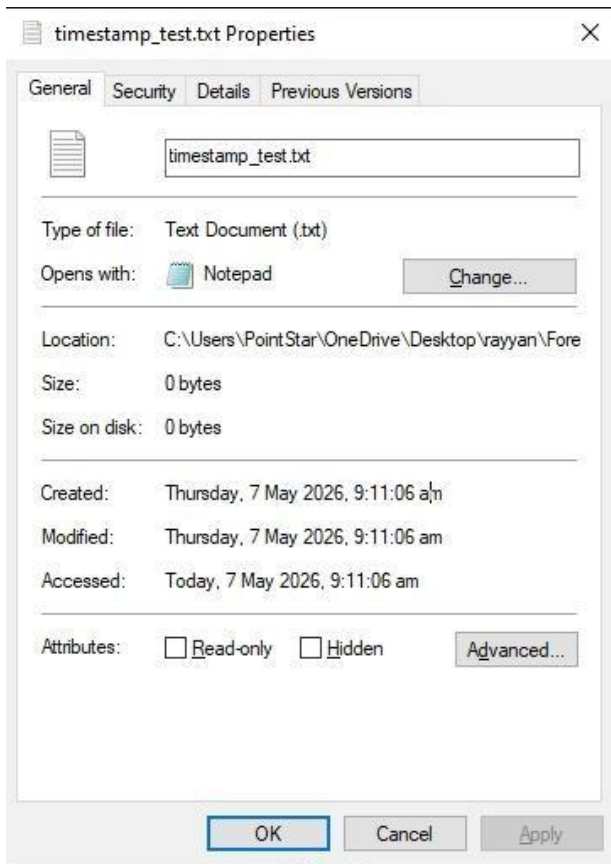
Cyberster Blue Team
Internship
WEEK 07 Report

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CSI-B1-487

Introduction

The focus of Week 7 was to transition from real-time monitoring to **Digital Forensics and Incident Response (DFIR)**. The primary objective was to perform a forensically sound acquisition of a storage device, conduct a deep-dive analysis of file system structures, and utilize automated tools for evidence triage and recovery.





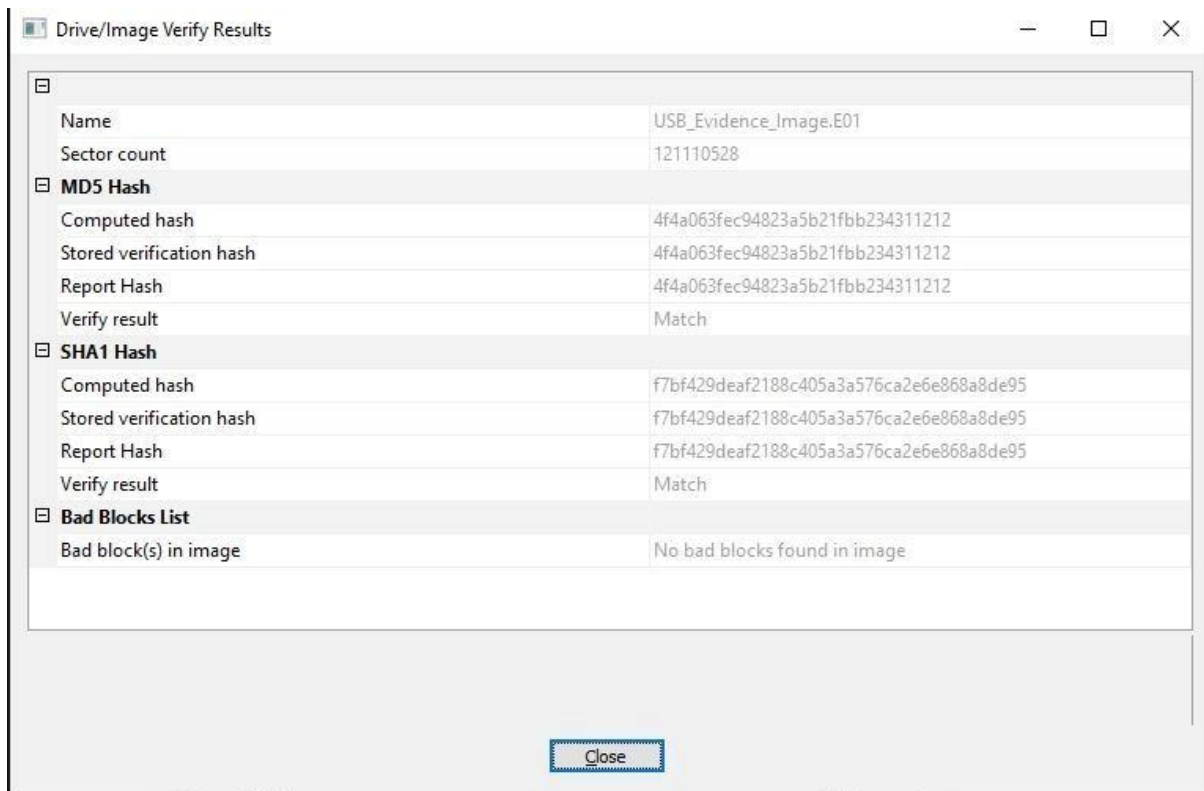
Day 46-47: Forensic Acquisition and Integrity Verification

To perform a bit-for-bit physical acquisition of a 64GB USB device while ensuring the integrity of the original evidence through hardware/software write-blocking.

- **Write-Blocking Implementation:** Prior to device connection, I implemented a software write-block via the Windows Registry (**WriteProtect = 1**) to prevent any metadata contamination by the host OS.
- **Imaging Methodology:** Using **FTK Imager v4.7.3.81**, I performed a **Physical Drive Acquisition** to capture the entire disk geometry, including unallocated and slack space.
- **Verification:** I generated an **E01 (EnCase)** image. Integrity was verified by generating MD5 and SHA1 hashes and comparing them against the original source.

Technical Proof:

- **MD5 Hash:** 4f4a063fec94823a5b21fbb234311212
- **Verification Result:** MATCH



Day 48-49: File System Analysis and Artifact Research

To analyze the structural characteristics of NTFS and demonstrate how data can be obfuscated or tracked through metadata.

- **Alternate Data Streams (ADS):** I demonstrated NTFS-specific data hiding by attaching a hidden text stream (`file.txt:hidden.txt`) to a host file. This remains invisible to standard directory listings but was revealed using forensic flags.
 - Logic: `dir /r` exposes the `$DATA` attribute of hidden streams.
- **MACB Timestamp Analysis:** I conducted experiments on file temporal data (Modified, Accessed, Changed, Birth). I documented that Copying a file

updates the Birth (B) timestamp while preserving the Modified (M) date—a key artifact for identifying external data introduction.

Technical Proof:

- **Command:** `dir /r`
- **Observation:** Identification of a 14-byte hidden stream on a file reporting 0 bytes.

```
07/05/2026 08:27 am <DIR> Downloads
11/08/2023 07:00 pm <DIR> Favorites
07/05/2026 09:06 am 0 file.txt
02/07/2025 01:12 pm <DIR> IdeaProjects
11/08/2023 07:00 pm <DIR> Links
```

```
11/08/2023 07:00 pm <DIR> Favorites
07/05/2026 09:06 am 0 file.txt
14 file.txt:hidden.txt:$DATA
02/07/2025 01:12 pm <DIR> IdeaProjects
```

Day 50-52: Forensic Triage and Evidence Recovery

To utilize the **Autopsy** forensic suite for automated triage, forensic carving, and evidence extraction.

- **Ingest Modules:** I configured modules for **File Type Identification**, **Extension Mismatch Detection**, and **PhotoRec Carver** to scan the E01 image.
- **Anomalous Detection:**
 - **Extension Mismatches (142):** Identified files where headers did not match extensions (e.g., a `.jpg` that is actually an executable).
 - **Encryption (40):** Flagged encrypted volumes for further analysis.
- **Evidence Extraction:** I successfully performed forensic carving in the **Deleted Files** view to recover and extract PDF documents that were no longer visible to the operating system.

Technical Proof:

Recovery: Successful extraction of UsersandRoles.pdf and other deleted artifacts.

The screenshot displays a forensic analysis interface with a sidebar on the left and a main window on the right. The sidebar contains a tree view with categories like Data Sources, File Views, Deleted Files, MB File Size, Data Artifacts, Analysis Results, OS Accounts, Tags, Score, and Reports. The 'Deleted Files' category is expanded, showing a list of files including 'File System (25312)' and 'All (27459)'. The main window shows a 'Listing' tab with a table of files. The table has columns for Name, S, C, O, Modified Time, Change Time, Access Time, and Created Time. The files listed include various MP3 files and a PDF file named 'UsersandRoles.pdf'. A dialog box is open in the center of the screen, displaying the message 'File(s) extracted.' with an 'OK' button.

File System

10000 Results

Page: 1 of 3 Pages: Go to Page: Save Table as CSV

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time
Aaj Sik Mitran Di Vadheri Aey.mp3				2017-06-18 13:26:50 PKT	0000-00-00 00:00:00	2024-01-20 11:33:28 PKT	2018-
Aaqa Teray Naam Dee Khatir.mp3				2015-05-12 05:31:34 PKT	0000-00-00 00:00:00	2024-01-20 11:33:28 PKT	2018-
Ae Sari Mahfil Mehk.mp3				2015-05-12 05:31:36 PKT	0000-00-00 00:00:00	2024-01-20 11:33:34 PKT	2018-
Aey Hassnain Kay Nana (Saifi Naem).mp3				2015-05-12 05:31:38 PKT	0000-00-00 00:00:00	2024-01-20 11:33:34 PKT	2018-
Aey Hassnain Kay Nana (Saifi Saif Ullah).mp3				2015-02-02 08:38:58 PKT	0000-00-00 00:00:00	2024-01-20 11:33:38 PKT	2018-
Aey Khould Key Baharoon .mp3				2015-07-12 05:07:44 PKT	0000-00-00 00:00:00	2024-01-20 11:33:38 PKT	2018-
Aey Meray Allah.mp3				2015-02-02 08:39:00 PKT	0000-00-00 00:00:00	2024-01-20 11:33:38 PKT	2018-
Ahsan Saifi.m4a				2017-07-03 12:01:46 PKT	0000-00-00 00:00:00	2024-01-20 11:33:40 PKT	2018-
Alif Allah.mp3				2016-02-20 05:33:32 PKT	0000-00-00 00:00:00	2024-01-20 11:33:40 PKT	2018-
Allah Allah Karda Dil Allah Allah Karda.mp3				2015-02-02 08:38:58 PKT	0000-00-00 00:00:00	2024-01-20 11:33:42 PKT	2018-

File(s) extracted.

OK

Displays tags and comments associated with the selected file.